MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 1	6.630 Electromagnetics	Issued: 090408R
Fall Term 2008		Due: 091108R

Reading assignment: Section 1.1, J. A. Kong,"*Electromagnetic Wave Theory*", EMW Publishing, 2008

Problem P1.1

Electromagnetic waves satisfy all of the Maxwell equations. Consider, in free space, the following electric field vectors:

$$\overline{E}_1 = \hat{z}\cos(\omega t - kz)$$
$$\overline{E}_2 = (\hat{x}+\hat{z})\cos\left(\omega t + k|x-z|/\sqrt{2}\right)$$
$$\overline{E}_3 = (\hat{x}+\hat{z})\cos(\omega t + ky)$$

(a) Do these electric field vectors satisfy the wave equation

$$\left(\nabla^2 - \mu_o\epsilon_o\frac{\partial^2}{\partial t^2}\right)\overline{E} = 0$$

What is the relationship between ω and k?

(b) Find the corresponding magnetic field vectors for each of the three electric field vectors.

(c) Which of the three fields qualify as electromagnetic waves? For those not qualified as electromagnetic waves, you should state which of the Maxwell equations are violated. Show that for those qualified as electromagnetic waves, the electric and magnetic field vectors are perpendicular to the direction of propagation.

Problem P1.2

The known spectrum of electromagnetic waves covers a wide range of frequencies. Electromagnetic phenomena are all described by Maxwell's equations and, by convention, are generally classified according to wavelengths or frequencies. Radio waves, television signals, radar beams, visible light, X rays, and gamma rays are examples of electromagnetic waves.

(a) Give in meters the wavelengths corresponding to the following frequencies:
- (i) 60 Hz
- (ii) AM radio (535–1605 kHz)
- (iii) FM radio (88–108 MHz)
- (iv) C- band (4–6 GHz)
- (v) Visible light ($\sim 10^{14}$ Hz)
- (vi) X-rays ($\sim 10^{18}$ Hz)

(b) Give in Hertz the temporal frequencies corresponding to the following wavelengths:
(i) 1 km, (ii) 1 m, (iii) 1 mm, (iv) 1 μm, (v) 1 Å.

(c) Give in unit K_o the spatial frequencies corresponding to the wavelengths in (b).

Problem P1.3

Prove the following vector identities, all of which are frequently used in this course.

$\nabla \times \left(\nabla \times \overline{E}\right) = \nabla \left(\nabla \cdot \overline{E}\right) - \nabla^2 \overline{E}$

$\nabla \cdot \left(\overline{E} \times \overline{H}\right) = \overline{H} \cdot \left(\nabla \times \overline{E}\right) - \overline{E} \cdot \left(\nabla \times \overline{H}\right)$

$\nabla \cdot \left(\nabla \times \overline{A}\right) = 0$

$\nabla \times (\nabla \phi) = 0$

Problem P1.4

Consider a wave which has the form of $\cos(kz - \omega t)$, where the phase term is $\phi = kz - \omega t$. Take derivative of the phase ϕ with respect to time t and let $\partial\phi/\partial t = 0$ to calculate the wave velocity. Can we calculate the wave velocity by taking derivative of the phase with respect to z? Why?

Run the following code in Matlab and you will get a figure showing the propagation of this wave. Trace the line where each point has the same phase as the origin. Then the slope of the line represents the wave velocity. Here we let $k = 1$ first. What will happen if k is increased?

```
k =1;
omega = 2;
[z,t] = meshgrid(0:0.01:10, 0:0.01:10);
wave = cos(k*z - omega*t);
pcolor(t,z,wave);shading flat;colorbar;
xlabel('t');ylabel('z');axis square
```

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 2	6.630 Electromagnetics	Issued: 091108R
Fall Term 2008		Due: 091808R

Reading assignment: Section 1.2-1.3 ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing.

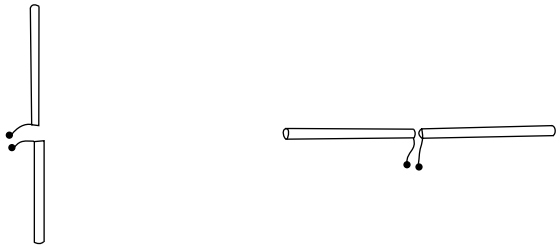
Problem P2.1

In the United States, broadcasting to the general public is regulated by the Federal Communications Commission (FCC), which allocates frequencies and establishes technical standards. Three general classes of broadcast stations have been established. *Standard broadcast stations* (amplitude modulation-AM) are licensed for operation on channels spaced by 10 kHz and occupying the band from 535 to 1605 kHz. *Frequency modulation* (FM) *broadcast stations* are authorized for operation on 100 allocated channels, each 200 kHz wide, extending consecutively from 201 on 88.1 MHz to channel 300 on 107.9 MHz. *Television broadcast stations* (operating with vestigial-sideband amplitude modulation of the visual carrier and frequency modulation of the aural carrier) are authorized for commercial and educational operation on designated channels 2–83, each 6 MHz wide, extending from 54–806 MHz. (See reference: *Reference Data for Radio Engineers*, Howard W. Sams & Co., ITT, 1983.)

The electromagnetic waves radiated by AM stations have the $\overline{E}$ field perpendicular to the ground and parallel to the antenna towers (vertical polarization). Most FM stations broadcast with circular polarization. For television broadcasting, the $\overline{E}$ field is parallel to the ground (horizontal polarization).

The induced current on a receiving antenna is largest when the antenna is aligned with the electric field. Given are two wire antenna configurations. For each type of broadcast, AM, FM, and TV, specify which configuration(s) gives maximum reception

(i) (ii)



Problem P2.2

Obtain the phasor notation of the following time-harmonic functions (if possible):

(a) $V(t) = 6\cos(\omega t + \pi/4)$
(b) $I(t) = -8\sin(\omega t)$
(c) $A(t) = 3\sin(\omega t) - 2\cos(\omega t)$
(d) $C(t) = 6\cos(120\pi t - \pi/2)$
(e) $D(t) = 1 - \cos(\omega t)$
(f) $U(t) = \sin(\omega t + \pi/3)\sin(\omega t + \pi/6)$

Problem P2.3

Consider an electromagnetic wave propagating in the $\hat{z}$-direction with

$$\overline{E} = \hat{x}e_x \cos(kz - \omega t + \psi_x) + \hat{y}e_y \cos(kz - \omega t + \psi_y)$$

where e_x, e_y, ψ_x, and ψ_y are all real numbers.

(a) Let $e_x = 2$, $e_y = 1$, $\psi_x = \pi/2$, $\psi_y = \pi/4$. What is the polarization?

(b) Let $e_x = 1$, $e_y = \psi_x = 0$. This is a linearly polarized wave. Prove that it can be expressed as the superposition of a right-hand circularly polarized wave and a left-hand circularly polarized wave. 叠加，重合

(c) Let $e_x = 1$, $\psi_x = \pi/4$, $\psi_y = -\pi/4$, $e_y = 1$. This is a circularly polarized wave. Prove that it can be decomposed into two linearly polarized waves.

Problem P2.4

Wave polarization can be viewed by either taking a series of still pictures at several fixed times, called the spatial view point or by making observations at a fixed point in space, called the temporal view point. We define polarization from the temporal view point. Let us now look at polarization from the spatial view point.

Consider an electromagnetic wave with $k = 100\mathrm{K_o}$ propagating in the $\hat{z}$ direction.

$$\overline{E}(\overline{r}, t) = E_0[\hat{x}\cos(kz - \omega t) + \hat{y}\sin(kz - \omega t)]$$

What is the wavelength and the polarization of this wave?

From the spatial point of view, by taking a picture at $t = 0$, the tips of the electric field vectors form a helix. Is the helix right-handed or left-handed? What is the pitch(wavelength) of this helix?

Problem P2.5

The Earth receives over all frequency bands about $1.5\,\mathrm{kW/m^2}$ of power from the Sun.

(a) The Earth-Sun distance is 150×10^9 m. How long does it take the sunlight to reach the Earth?

(b) The Earth radius is 6400 km. What is the total power received by the Earth?

(c) Assume the Sun's mass is $2 \times 10^{30}\,\mathrm{kg}$ which converts to radiated energy according to mc^2 at 1 percent efficiency. How long can the Sun radiate at the present level?

(c) The Sun radiates $10^{-20}\ \mathrm{W\,m^{-2}\,Hz^{-1}}$ at 3 GHz. Assuming constant power level over 1 GHz bandwidth, what is the Poynting power density and the corresponding electric field amplitude?

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
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Problem Set No. 5	6.630 Electromagnetics	Issued: 100208R
Fall Term 2008		Due: 100908R

Reading assignment: Section 2.2, 2.3 ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P5.1

Digital systems on printed circuit boards often incorporate source terminations instead of load terminations to minimize the dissipated power and to reduce the magnitude of propagating voltage waveforms. Consider the digital system in the figure. The driver is modeled by a voltage source $V_s = 1$ V and a source resistance $R_s = 25\ \Omega$. The transmission line has a characteristic impedance $Z_0 = 100\ \Omega$ and a propagation velocity $v = c/2 = 1.5 \cdot 10^8$ m/s due to the geometry and dielectric constant of the the printed circuit board. Digital receivers often have very low input capacitance so that the load can be modeled as an open circuit in the frequency range of interest. The source termination is implemented by inserting a series resistance $R_1 = 75\ \Omega$ so that the total input resistance matches the characteristic impedance of the transmission line.

(a) The length of the transmission line is measured to be 30 cm. Calculate the round trip propagation time for a signal transient.

(b) A low to high transition is modeled by a switch that closes at $t = 0$. If you measure the voltage at V_1, you might conclude that the quality of the signal is poor. Plot the voltage at V_1 versus time assuming zero volts initially on the line.

(c) Now assume that the signal quality measurement is made at the digital receiver. Plot the voltage V_2 versus time assuming zero volts initially on the line.

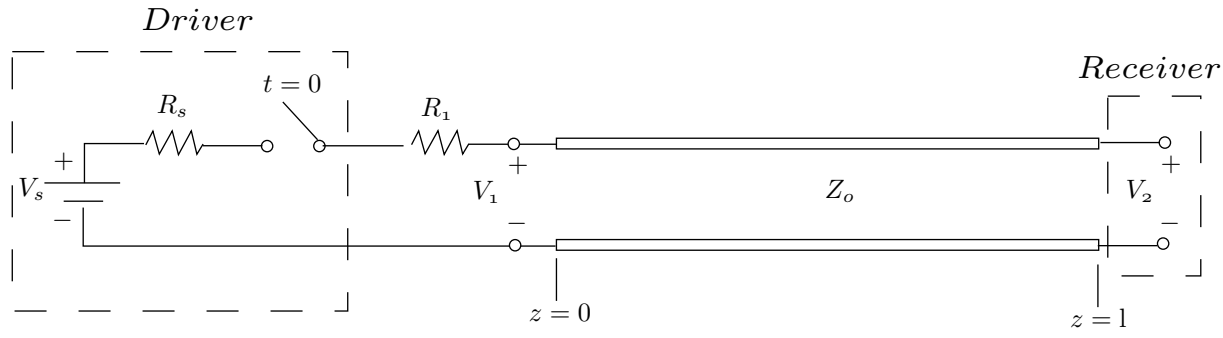


Fig. 1

Problem P5.2

Consider a TEM transmission line as shown in the following figure. The characteristic impedance of the transmission line is Z_0 and its length is $l = 2\lambda$, where λ is the wavelength of the wave in the line. The load impedance is Z_L. The current on the line is given by

$$I(z) = I_o \cos kz.$$

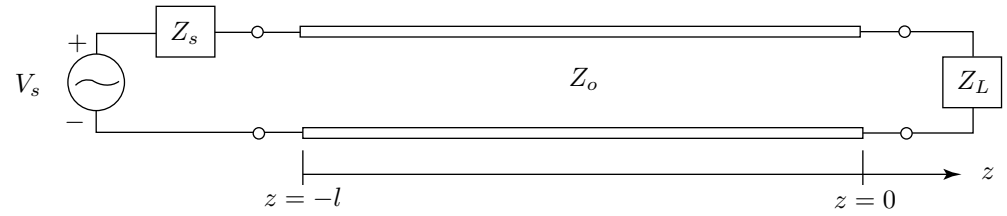


Fig. 2

(a) What is V_+ and what is V_- in terms of Z_o and I_o?
(b) What is the reflection coefficient at the load Γ_L?
(c) Show that the load impedance $Z_L = 0$.
(d) The real voltage in space and time is defined as $V(z,t) = \mathrm{Re}\{V(z)e^{j\omega t}\}$. Let I_o and Z_o be real, write down the expression and sketch the voltage on the line at $\omega t = \pi/2$.
(e) Let the voltage of the source be $V_s = I_o Z_o$, what is the source impedance Z_s in terms of Z_o?

Problem P5.3

Consider the transmission line system which is composed of three parts $T1$, $T2$ and $T3$ as shown in Fig. 3. All of them have the same length ℓ. $T1$ and $T2$ have the characteristic impedance Z_o while $T3$ has the characteristic impedance $2Z_o$. $T2$ is open at the right end while $T3$ is short at the right end.

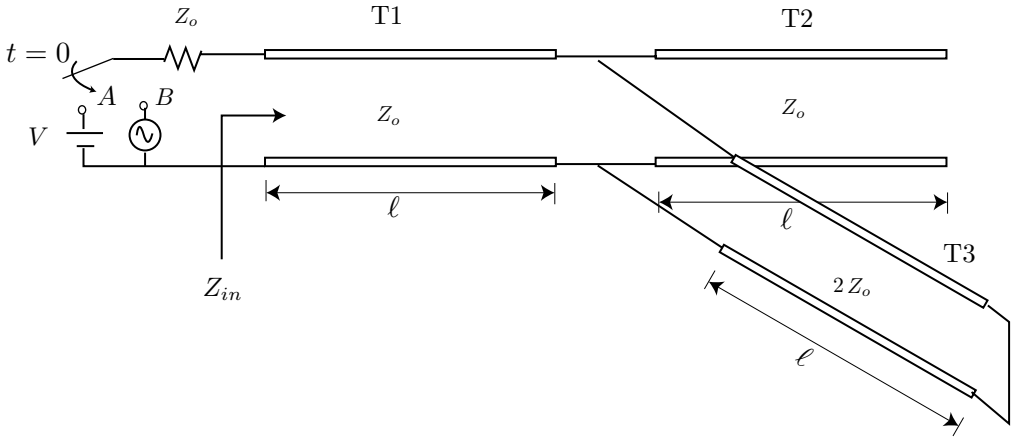


Fig. 3

(a) If at time $t = 0$ the switch is connected to A, sketch the voltage form $V(z)$ on $T1$ at the time $t = 1.5\ell/v$. Here v is the speed of the wave.
(b) Now the switch is connected to B and after a long time the sinusoidal steady state has been built up. Calculate the input impedance Z_{in} at the input end. ($k\ell = \pi$.)

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Problem Set No. 3	6.630 Electromagnetics	Issued: 091808R
Fall Term 2008		Due: 092508R

Reading assignment: Section 1.3, 1.5, 3.1 ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P3.1

According to the Bohr model of an atom, electrons revolve around the nucleus in quantized orbits with radii $R = n\hbar/mv$ where n is an integer, m is the electron mass and v is the electron velocity. Letting the nucleus be a positive charge of Ze, calculate R by equating the centrifugal force with the Lorentz force. Estimate the radius for a hydrogen atom with $Z = 1$.

Problem P3.2

(a) At the operating frequency (2.5 GHz) of a microwave oven, the permittivity for bottom round steak is about $\epsilon = 40\epsilon_0$ and the conductivity $\sigma = 2$ mho/meter. What is the penetration depth? Compare this penetration depth to that of polystyrene foam which has the permittivity $\epsilon = 1.03\epsilon_0$ and conductivity $\sigma = 4 \times 10^{-6}$ mho/meter.

(b) Earth is considered to be a good conductor when $\omega\epsilon/\sigma \ll 1$. Determine the highest frequency for which earth can be considered a good conductor if $\ll 1$ means less than 0.1. Assume $\sigma = 5 \times 10^{-3}$ mho/meter and $\epsilon = 10\epsilon_0$.

(c) Aluminum has $\epsilon = \epsilon_0$, $\mu = \mu_0$ and $\sigma = 3.54 \times 10^7$ mho/m. If an antenna for VHF reception is made of wood coated with a layer of aluminum and if its thickness ought to be five times greater than the skin depth of the aluminum at that frequency, determine the thickness of the aluminum layer. Is ordinary aluminum foil thick enough for that purpose? Use $f = 100$ MHz. Ordinary aluminum is approximately 1/1000 inch thick.

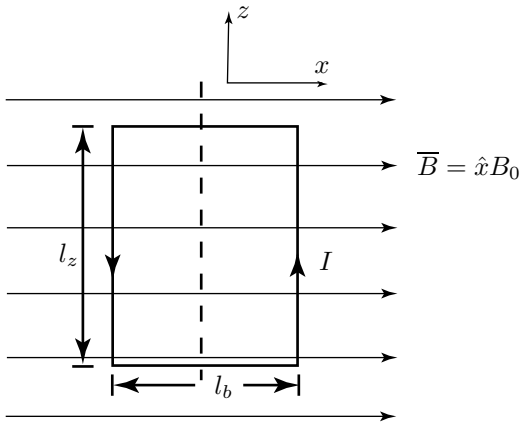
(d) Calculate skin depths for sea water at frequencies 100 Hz and 5 MHz. Sea water can be characterized by conductivity $\sigma = 4$ mho/m, permittivity $\epsilon = 80\epsilon_0$, and permeability $\mu = \mu_0$ at those frequencies.

(e) A ship at the ocean surface wishes to communicate electromagnetically with a deeply submerged vehicle 100 meters below the surface. Consider a ULF signal at 1 KHz propagating down into the sea water. What fraction of the incident power density reaches the submerged vehicle?

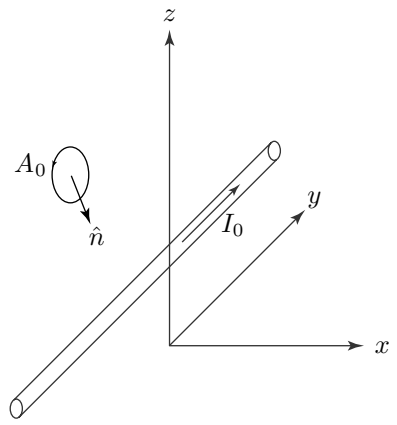
Problem P3.3

The magnetic moment $\overline{M}$ is analogous to the expression for the mechnical angular momentum. We will show that the magnetic torque can be expressed by $\overline{T} = \overline{M} \times \overline{B}$.

(a) As shown in the figure below, a current loop is placed in a uniform magnetic field. Given $\overline{T} = \overline{r} \times \overline{F}$, use Lorentz force law to show that $\overline{T} = \hat{z}IAB_0$. By writing $\overline{M} = -\hat{y}IA$ and $\overline{B} = \hat{x}B_0$, show that $\overline{T} = \overline{M} \times \overline{B}$.



(b) As shown in the figure below, a loop carrying a current of I_l with normal $\hat{n} = \frac{1}{\sqrt{2}}(\hat{x} - \hat{y})$ is placed a distance d above a straight wire, which is carrying a current of I_0. Calculate the magnetic moment of the current loop and the magnetic field generated by straight wire at the loop's position. Using these two values, calculate the torque vector, $\overline{T}$, of loop. In what direction does the loop move due to the torque?



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Problem Set No. 2	6.632 Electromagnetic Wave Theory	Issued: 02/15/07 R
Spring Term 2007		Due: 02/22/07 R

Reading assignment: Section 3.2 J. A. Kong, "*Electromagnetic Wave Theory*"

Problem P2.1

(a) The complex permittivity for bottom round steak is about $\epsilon = 40(1 + i0.3)\epsilon_o$ at the operating frequency (2.5 GHz) of a microwave oven. What is the penetration depth?

(b) Calculate loss tangents and skin depths for sea water at frequencies 60 Hz and 10 MHz. Sea water can be characterized by conductivity $\sigma = 4\,\text{mho/m}$, permittivity $\epsilon = 80\epsilon_0$, and permeability $\mu = \mu_o$ at those frequencies.

(c) A 100-Hz electromagnetic wave is propagating down into the sea water with an electric field intensity E of 1 V/m just beneath the sea surface. What is the intensity of E at a depth of 100 m? What are the time-average Poynting's power densities just beneath the surface and at a depth of 100 m?

Problem P2.2

Superconductivity was first observed by Kamerlingh Onnes in 1911. In 1933 Meissner and Ochsenfeld discovered that superconducting metals cannot be penetrated by magnetic fields. Magnetic fields are expelled from a normal metal when it is cooled to the superconducting state. The macroscopic theory of superconductivity was developed by London and London in 1935 followed by the microscopic theory of Bardeen, Cooper and Schrieffer in 1957.

A simple model of superconductivity calls for an electron plasma with a very high electron density N.

(a) Show that the penetration depth of a plasma with very large N takes the form

$$d_p = \sqrt{\frac{m}{Ne^2\mu_0}}$$

(b) Letting $N = 7 \times 10^{28}\,\text{m}^{-3}$, calculate d_p.

(c) Compare the above result with the skin depth of a good conductor. Explain why a very slowly varying magnetic field can penetrate a good conductor but not a superconductor.

Problem P2.3

An electron plasma becomes anisotropic when an external dc magnetic field $\overline{B}_0$ is applied. Assume that $\overline{B}_0$ is in the direction. The magnetic field enters the constitutive parameters via the cyclotron frequency $\omega_c = qB_0/m$. Show that the permittivity tensor is

$$\overline{\overline{\epsilon}} = \begin{pmatrix} \epsilon & i\epsilon_g & 0 \\ -i\epsilon_g & \epsilon & 0 \\ 0 & 0 & \epsilon_z \end{pmatrix}$$

with the constitutive parameters defined by

$$\epsilon = \epsilon_0 \left[1 - \frac{\omega_p^2}{(\omega^2 - \omega_c^2)} \right]$$
$$\epsilon_g = \epsilon_0 \left[\frac{-\omega_p^2 \omega_c}{\omega(\omega^2 - \omega_c^2)} \right]$$
$$\epsilon_z = \epsilon_0 \left[1 - \frac{\omega_p^2}{\omega^2} \right]$$

Write the constitutive relation as $\overline{E} = \overline{\overline{\kappa}} \cdot \overline{D}$ with

$$\overline{\overline{\kappa}} = \overline{\overline{\epsilon}}^{-1} = \begin{pmatrix} \kappa & i\kappa_g & 0 \\ -i\kappa_g & \kappa & 0 \\ 0 & 0 & \kappa_z \end{pmatrix}$$

Determine the constitutive parameters κ, κ_g, and κ_z.

Problem P2.4

Show that a linearly polarized wave can be decomposed into a right-hand circularly polarized wave and a left-hand circularly polarized wave.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 4	6.630 Electromagnetics	Issued: 092508R
Fall Term 2008		Due: 100208R

Reading assignment: Section 3.1, 3.3, 2,1, 2.2 ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P4.1

Consider a conductive uniaxial medium with

$$\overline{\overline{\epsilon}} = \begin{pmatrix} \epsilon & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon_z \end{pmatrix}$$

and

$$\overline{\overline{\sigma}} = \begin{pmatrix} \sigma & 0 & 0 \\ 0 & \sigma & 0 \\ 0 & 0 & \sigma_z \end{pmatrix}$$

Find dispersion relations for this medium when a plane wave is propagating in $\hat{x}$ direction. Explain the operation of a polaroid with this model by assuming $\sigma_z/\sigma \gg 1$. Show that a piece of polaroid turns any wave into a linearly polarized wave.

Problem P4.2

Consider a circularly polarized electromagnetic wave normally incident upon a slab as shown in the figure. The incident electric field is expressed by

$$\overline{E}_i = E_o\hat{x}e^{ik_y y} + \alpha E_o\hat{z}e^{i(k_y y+\beta)}.$$

In this problem, neglect the reflection of the slab.

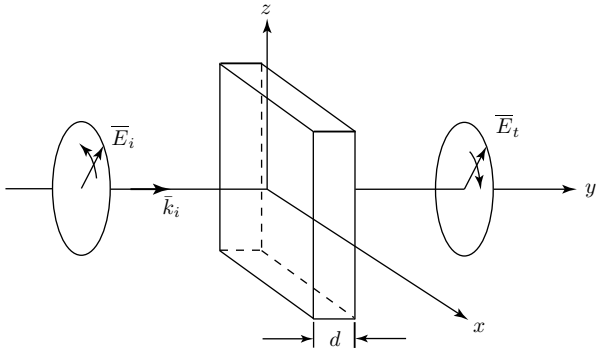


Fig. 1

(a) Let the incident wave be left-hand circularly polarized and assume that both α and β are positive, what is α and what is β?

(b) Let the slab be a uniaxial medium with the permittivity tensor

$$\overline{\overline{\epsilon}} = \begin{bmatrix} \epsilon_x & 0 & 0 \\ 0 & \epsilon_y & 0 \\ 0 & 0 & \epsilon_z \end{bmatrix}$$

where $\epsilon_x = \epsilon_y = 4\epsilon_o$, $\epsilon_z = 9\epsilon_o$ and the permeability $\mu = \mu_o$. Inside the uniaxial slab, what is the wave number k_y for the $\hat{x}$-polarized electric wave in terms of the wave number in free space k_o, where $k_o = \omega\sqrt{\mu_o\epsilon_o}$?

(c) For the uniaxial slab as in Part (b), let the incident wave be left-hand circularly polarized, what is the minimum thickness d in terms of the wavelength in free space λ_o, where $\lambda_o = 2\pi/k_o = 2\pi/\omega\sqrt{\mu_o\epsilon_o}$, such that the output electric field is right-hand circularly polarized?

Problem P4.3

A break in a high-voltage DC power line occurs at $z = 0$ at time $t = 0$, as shown in Fig.2. The line was carrying a DC voltage V_o and DC current I_o before the break occurred. Assume that the tree is non-conducting.

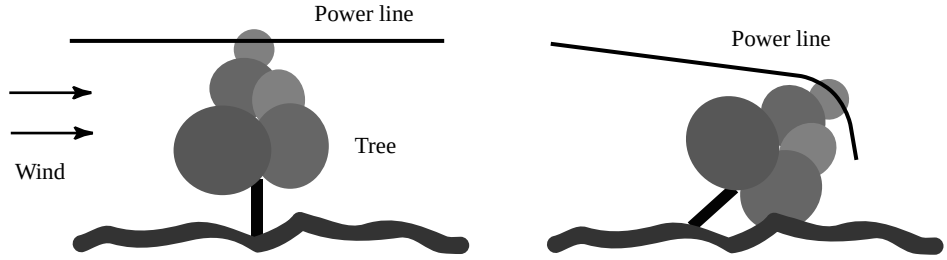


Fig. 2

(a) Sketch I and V on the line at some time t after the break has occurred, but before any reflections from the source and load ends. The characteristic impedance of the line is Z_0.

(b) Consider a 600 kV line, carrying a power of 10^3 megawatts, with a characteristic impedance of $500\,\Omega$ (two-wire line). What is the peak voltage on the line after the break occurs?

Problem P4.4

A very long transmission line with characteristic impedance Z_0 and wave velocity $v = c$ has a shunt resistor of unknown vlue R_L at an unknown location $z = \ell$. A measurement of the voltage at the input, $V_0(t)$, with a unit step generator applied to the line, yields the result shown in Fig. 3.

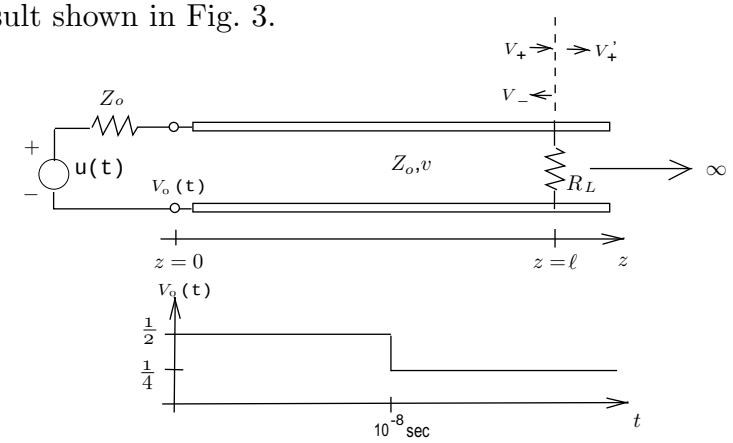


Fig. 3

(a) What is ℓ?

(b) What is R_L?

(c) Sketch the voltage and current distribution on the line at the time $t = 1.5\ell/v$.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 2	6.632 Electromagnetic Wave Theory	Issued: 20090212 R
Spring Term 2009		Due: 20090219 R

Reading assignment: Section 3.3 ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P2.1

In a negative uniaxial medium ($\epsilon_z < \epsilon$), a wave vector $\overline{k}$ makes an angle $\theta_1 < \pi/2$ with the optic axis. The Poynting's vector $< \overline{S} >$ makes an angle θ_2 with the optic axis. Determine whether θ_2 is larger or smaller than θ_1. Sketch the curve of θ_2 vs θ_1 for $0 < \theta_1 < \pi/2$.

Problem P2.2

Consider a slab of thickness d, as shown in Fig. 1, with the following permittivity and conductivity tensors:

$$\overline{\overline{\epsilon}} = \begin{bmatrix} \epsilon_x & 0 & 0 \\ 0 & \epsilon_y & 0 \\ 0 & 0 & \epsilon_z \end{bmatrix} \qquad \overline{\overline{\sigma}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma_z \end{bmatrix}$$

where $\epsilon_x = 12\epsilon_\circ$, $\epsilon_y = \epsilon_\circ$, $\epsilon_z = 4\epsilon_\circ$, and $\mu = \mu_\circ$. The conductivity for polarized wave in z-direction is $\sigma_z = 0.2\epsilon_\circ\omega$ mho/meter.

Let the incident electric field propagate in the $\hat{w}$-direction. Neglect reflections.

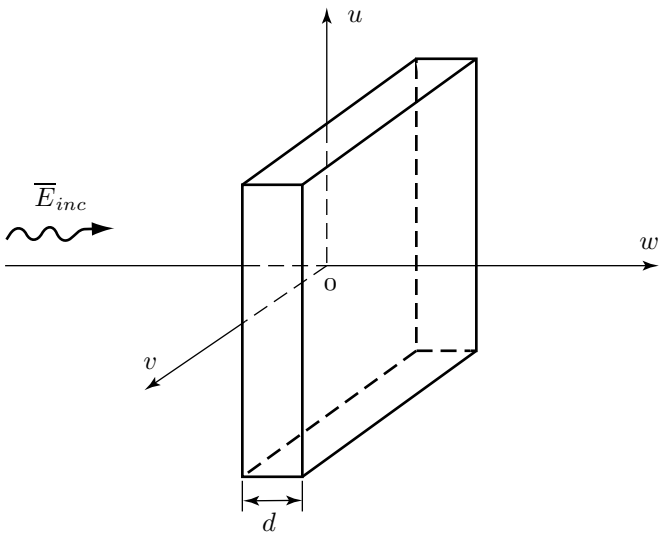


Fig. 1

Polarizer:

(a) Assign x, y, z to u, v, w (not necessarily in that order) so that, for any arbitrarily-polarized incident electric field and sufficiently thick slab, the transmitted field is linearly polarized.

(b) Determine the minimum thickness d in terms of free space wavelength such that the undesirable component of the incident field is attenuated by $1/e$.

Wave plate:

(c) Assign x, y, z to u, v, w (not necessarily in that order) so that, for a given linearly polarized incident electric field, the transmitted field is circularly polarized. Specify the axes so that there is no power absorption. Give an expression for an incident electric field such that, given the correct thickness d, the transmitted electric field is circularly polarized. What is the minimum correct d? Sketch the rotation of E field on the right surface (i. e. $w = d$) of the slab with this minimum d.

Problem P2.3

In a ferrite, the magnetic moment $\overline{M}$ roughly obeys the relationship $d\overline{M}/dt = g\mu_0\overline{M} \times \overline{H}$, where g is the gyromagnetic ratio. When a $\hat{z}$-directed dc magnetic field $\overline{H}_0$ (zeroth order) is present, the total fields take the form $\overline{H} = \hat{z}H_0 + \overline{H}_1$, $\overline{M} = \hat{z}M_0 + \overline{M}_1$, and $\overline{B} = \mu_0(\overline{H} + \overline{M})$.

(a) For the cross product $\hat{z} \times \overline{H}_1$, convert it to be $\overline{\overline{z}} \cdot \overline{H}_1$. What is $\overline{\overline{z}}$?

(b) To the first order approximation, $\overline{M}_1 \times \overline{H}_1$ is negeligible. Show that the permeability $\overline{\overline{\mu}}$ which satisfies $\overline{B}_1 = \overline{\overline{\mu}} \cdot \overline{H}_1$ has the form of

$$\overline{\overline{\mu}} = \begin{bmatrix} \mu & i\mu_g & 0 \\ -i\mu_g & \mu & 0 \\ 0 & 0 & \mu_z \end{bmatrix}$$

(c) Find dispersion relations for the first-order fields in kDB system.

(d) Show that Faraday rotation exists in the ferrite.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 3 6.632 Electromagnetic Wave Theory Issued: 02/21/2008 R
Spring Term 2008 Due: 02/28/2008 R

Reading assignment: Section 3.3, 3.4 ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2005.

Problem P3.1

Use the *kDB* system to determine the dispersion relations for a biisotropic medium (Tellegen medium) with the constitutive relation

$$\overline{D} = \epsilon\overline{E} + \xi\overline{H}$$
$$\overline{B} = \xi\overline{E} + \mu\overline{H}$$

Problem P3.2

A Nicol prism made of calcite is cut diagonally and then joined together with a film of Canada balsam (refractive index n=1.53). Calcite is a negative uniaxial crystal with $\sqrt{\epsilon_z/\epsilon} = 1.49/1.66$. An incident light from the left will become a linearly polarized light when it leaves the crystal from the right. Show that with the arrangement shown in Fig. 1 an incident light from the left becomes a linearly polarized light when it leaves the crystal from the right. ($\alpha \approx 30.5^\circ$)

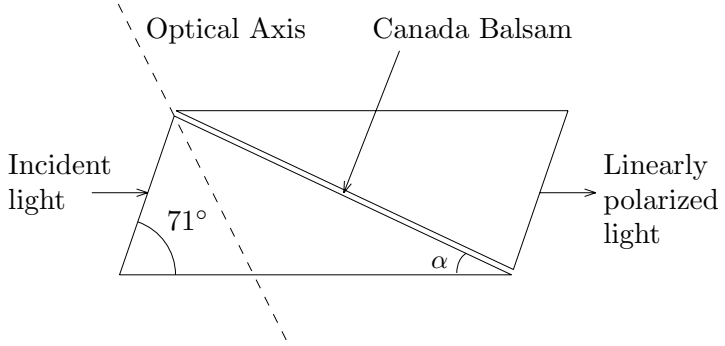


Fig. 1.

Problem P3.3

In a ferrite, the magnetic moment $\overline{M}$ roughly obeys the relationship $d\overline{M}/dt = g\mu_0\overline{M}\times\overline{H}$, where g is the gyromagnetic ratio. When a $\hat{z}$-directed dc magnetic field $\overline{H}_0$ (zeroth order) is present, the total fields take the form $\overline{H} = \hat{z}H_0 + \overline{H}_1$, $\overline{M} = \hat{z}M_0 + \overline{M}_1$, and $\overline{B} = \mu_0(\overline{H} + \overline{M})$. Find dispersion relations for the first-order fields. Show that Faraday rotation exists in the ferrite.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 5 | 6.632 Electromagnetic Wave Theory | Issued: 03/08/07 R
Spring Term 2007 | | Due: 03/15/07 R

4.1-1.3

Reading assignment: Section ~~3.3, 3.4~~ J. A. Kong, "*Electromagnetic Wave Theory*"
Quiz 1: March 22, 3:00 - 5:00 P.M., Rm. 26-314

Problem P5.1

Consider an electromagnetic wave impinging normally upon a dielectric half space (Region 2) with permittivity ϵ_2 from a medium (Region 1) with permittivity ϵ_1.

(a) Let $\epsilon_1 = \epsilon_o$ and $\epsilon_2 = 4\epsilon_o$. What are the reflection coefficient R_{12} and the transmission coefficient T_{12}?
(b) What is the sum of Poynting power of the wave on either side of the interface? Do they conserve?
(c) What is the sum of momentum density of the wave on either side of the interface? Do they conserve?
(d) Find the radiation pressure exerted on both sides of the boundary. Do they match?
(e) Will the half space move towards the incident wave or away from it?

Problem P5.2

The constitutive relation for a lossy dielectric with $\epsilon = \epsilon_R + i\epsilon_I$ and $\mu = \mu_0$ can be written as a material contribution in excess of the "free space" part $\epsilon_0\bar{E}$ by defining the polarization $\bar{P} = \bar{D} - \epsilon_0\bar{E}$.

(a) Show that the material contribution in maxwell equations can be contained within an effective electric current density $\bar{J}_e = -i\omega\bar{P}$.
(b) In a conductor, a conduction current $\bar{J}_c = \sigma\bar{E}$ is responsible for the loss of electromagnetic energy. More generally, we can write $\bar{J}_c = \omega\epsilon_I\bar{E}$. If we define the free current density by $\bar{J}_c$ and a bound current density $\bar{J}_b$ such that $\bar{J}_e \equiv \bar{J}_c + \bar{J}_b$, what is $\bar{J}_b$?

Now consider an electromagnetic wave $\bar{E} = \hat{x}E_0e^{-k_Iz}e^{ik_Rz}$ propagating in a lossy dielectric with $k = k_R + ik_I = \omega\sqrt{\mu_0\epsilon}$.

(c) What is the magnetic field of the electromagnetic wave?
(d) Show that the time-average Lorentz force density $< \bar{f}_b >= \frac{1}{2}\mathrm{Re}\{\bar{J}_b \times \bar{B}^*\}$ on bound currents reduces to

$$< \bar{f}_b >= -\hat{z}\frac{1}{2}k_{zI}(\epsilon_R - \epsilon_0)|E|^2,$$

where the negative sign indicates that the force on bound currents is opposite to the incident wave propagation direction when $\epsilon_R > \epsilon_0$.

This radiation attraction was observed by Poynting and Barlow in 1910 when an absorbing disk was initially attracted toward an incident light due to the heating and expulsion of occluded gas from the unilluminated side of the disk. To understand why the gas gained a momentum larger than the incident light, it is necessary to evaluate $< \bar{f}_c >$.

(e) Show that the time-average Lorentz force density on free currents is

$$< \bar{f}_c >= \hat{z}\frac{1}{2}k_R\epsilon_I|E|^2 = -\frac{1}{2}\mathrm{Re}\left\{\frac{n}{c}\nabla\cdot\bar{S}\right\},$$

where $n = ck_R/\omega$ is the index of refraction and $\bar{S} = \bar{E}\times\bar{H}^*$ is the complex Poynting vector resulting from the application of Poynting's theorem to the second equality.

Thus, the momentum transfer to free currents due to absorption of electromagnetic energy is proportional to the index of refraction. This dependence upon n has been observed in the Photon drag measurements [Gibson et. al., 1980] and the absorption of photons in a dilute gas of atoms [Campbell et. al., 2005].

Problem P5.3

A plane wave of angular frequency ω is incident on a plasma medium with permeability μ_o and permittivity $\epsilon = \epsilon_o\left(1-\omega_p^2/\omega^2\right)$, where ω_p is the plasma frequency and $\omega = 2\omega_p$.

(a) Calculate the critical angle θ_C such that the incident wave is totally reflected.

(b) Calculate the Brewster angle θ_B such that TM waves are totally transmitted.

(c) In general for any two isotropic media, can you find an incident angle θ such that $\theta = \theta_B > \theta_C$? If you can, give an example. If you cannot, explain why not.

Problem P5.4

The ionosphere extends from approximately 50 km above the earth to several earth radii (mean earth radius is about 6371 km) with the maximum in ionization density at about 300 km. For simplicity, assume that the ionosphere consists of a 40 km thick E layer with electron density $N = 10^{11}\,\mathrm{m}^{-3}$ below a 200 km thick F layer with $N = 6\times10^{11}\,\mathrm{m}^{-3}$.

(a) What are the plasma frequencies of the E and F layers ?

(b) Consider a plane wave of 10 MHz incident at an angle θ upon the ionosphere from below the E layer, what is the angle θ_t of the wave in the ionospheric E layer ?

(c) Let $\theta = 30^\circ$, below what frequency will the wave be totally reflected by the E layer and below what frequency will it be totally reflected by the F layer ?

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Problem Set No. 7	6.630 Electromagnetics	Issued: 20081023R
Fall Term 2008		Due: 20081030R

Reading assignment: Section 1.7 A-C, 4.1 A-C ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P7.1

A plane wave in free space is incident at an angle θ on a conducting half-space. For large $\sigma/\omega\epsilon_o$, show that the transmitted wave is almost perpendicular to the boundary by finding $\theta_t = \tan^{-1}(k_x/k_{zR})$.

Problem P7.2

When the incident k vector is normal to a plane boundary, a TE wave becomes a TEM wave; a TM wave also becomes a TEM wave. Compare the reflection and transmission coefficients for TE and TM waves at normal incidence. Do both TE and TM results reduce to the same number? If not, why? Do the reflectivities and transmissivities for TE and TM waves at normal incidence reduce to the same result?

Problem P7.3

A plane wave is incident from free space ($z > 0$) on a perfect electric conductor ($z < 0$) with the incident electric field

$$\overline{E}_i = (\frac{1}{\sqrt{2}}\hat{x} + \frac{1}{\sqrt{2}}\hat{z} + i\hat{y})e^{i\frac{1}{\sqrt{2}}k_o x - i\frac{1}{\sqrt{2}}k_o z}.$$

What is the reflected electric field $\overline{E}_r$? Compare the handedness of polarizations of $\overline{E}_i$ and $\overline{E}_r$.

Problem P7.4

Rainbow arc often appears when sunlight shines on water droplets after a brief shower late in the afternoon. When a sun ray is refracted as it enters the raindrop, total internally reflected from inside the drop, and refracted again as it leaves the drop and passes to the observer.

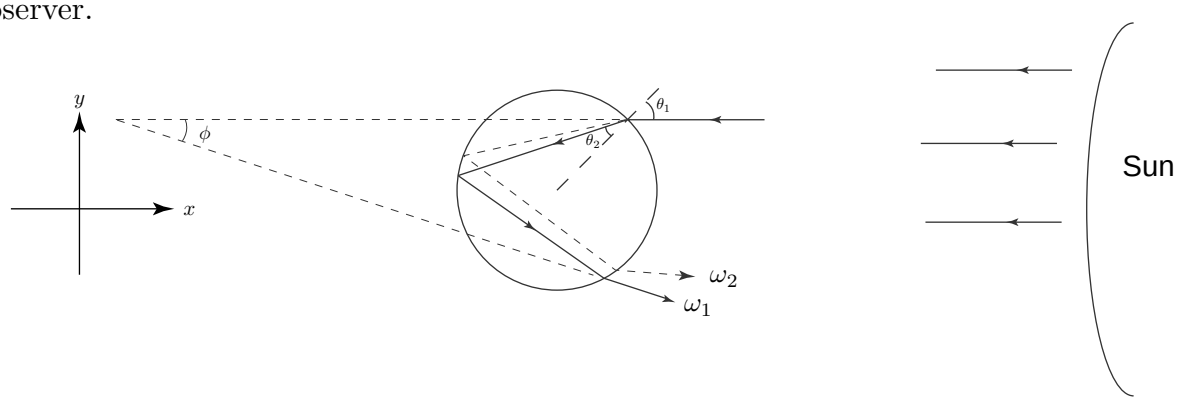


Fig. 1

(a) Consider the ray path with only one internal reflection. Show that the scattering angle ϕ between the incident ray and the exit ray is $2(2\theta_2 - \theta_1)$, where θ_1 is the incident angle and θ_2 is the refracted angle.

(b) For a sphere with a radius a where $ka \gg 1$, the direction where the scattering angle is stationary ($\frac{d\phi}{d\theta_1} = 0$) corresponds to the least cancellation between different rays and hence a large scattering amplitude. Show that the maximum scattering angle ($\phi_{\max}$) occurs at $\theta_1 = \sin^{-1}\sqrt{(4-n^2)/3}$ and $\phi_{max} \approx 42^\circ$ for $n = 4/3$, with the scattering angle between the incident ray and scattered ray $\theta_s = 138^\circ$.

(c) The refractive index for a raindrop is $n = 1.330$ for red light ($\lambda = 0.7\mu\text{m}$), $n = 4/3 = 1.333$ for orange light, and $n = 1.342$ for violet light ($\lambda = 0.4\mu\text{m}$). Determine the scattering angles for the red and violet light rays. What are the relative positions of the different color bands in a rainbow?

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
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Problem Set No. 9	6.630 Electromagnetics	Issued: 20081106R
Fall Term 2008		Due: 20081113R

Reading assignment: Section 4.2 A-C, G ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P9.1

What is the lowest frequency of an electromagnetic wave that can be propagating in the TE mode in the earth-ionosphere waveguide? Model the latter as two perfectly conducting parallel plates separated by 80 km.

Problem P9.2

Consider a perfectly conducting parallel-plate waveguide filled with a dielectric medium for $z > 0$ as shown in the figure. The dielectric medium has permittivity ϵ_1 . The operating frequency is 30 GHz. The guided wave propagates in the $\hat{z}$ direction.

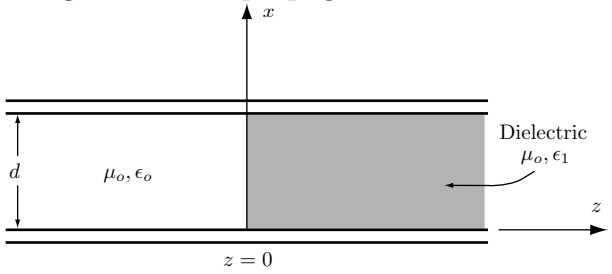


Fig. 1.

(a) Let $d = \sqrt{3}$ cm and consider the empty waveguide with $\epsilon_1 = \epsilon_o$ (in the absence of the dielectric). Which TE_m and TM_m modes can propagate in this waveguide?

(b) Find expressions for the $\overline{E}$ and $\overline{H}$ fields for the TM_2 mode in the absence of the dielectric.

(c) What are the phase and group velocities for the TM_2 mode at this operating frequency in the absence of the dielectric?

(d) Let $\epsilon_1 = 3\epsilon_o$ and $d = \sqrt{3}$ cm . For waves propagating in the $+\hat{z}$ direction, for which values of m will the TM_m modes be totally reflected at the dielectric boundary? Why?

(e) Let $\epsilon_1 = 3\epsilon_o$ and $d = \sqrt{3}$ cm . Calculate k_x and k_z for the TM_m mode. Will the TM_m mode be totally transmitted (no reflection) and why?

Problem P9.3

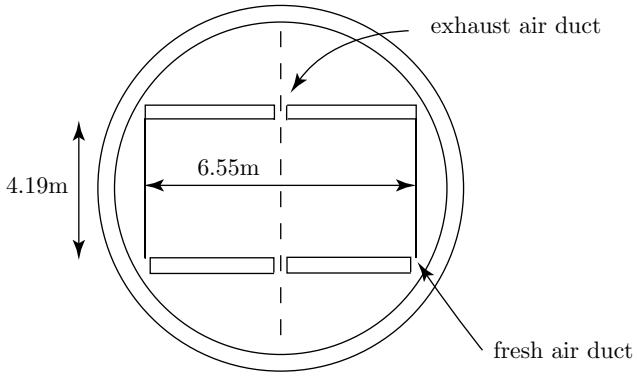
Practical rectangular-geometry optical waveguides are most easily understood as variants of the simple dielectric slab waveguide. Consider a plane slab of polystyrene ($\epsilon = 2.56\epsilon_o$) with 1 cm thickness whose operation frequency is 30 GHz. How many propogating modes are available using this dielectric waveguide? Determine the operation frequency range within which only the fundamental mode(s) is(are) available.

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Problem Set No. 10	6.630 Electromagnetics	Issued: 20081113R
Fall Term 2008		Due: 20081120R

Note: Quiz 2 will be held on Tuesday, Dec. 9, from 2:30-4pm in Rm 32-124.
Reading assignment: Section 4.2 C, H, I ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P10.1



Tunnel modeled as rectangular waveguide.

An AM(535-1605 kHz) radio in an automobile cannot receive any signal when the car is inside a tunnel. Consider, for example, the Lincoln Tunnel under the Hudson River, which was built in 1939. A cross-section of the tunnel as shown in the figure. Ignore the air ducts; assume they are closed. Model the tunnel as a rectangular waveguide of dimension $6.55m \times 4.19m$.

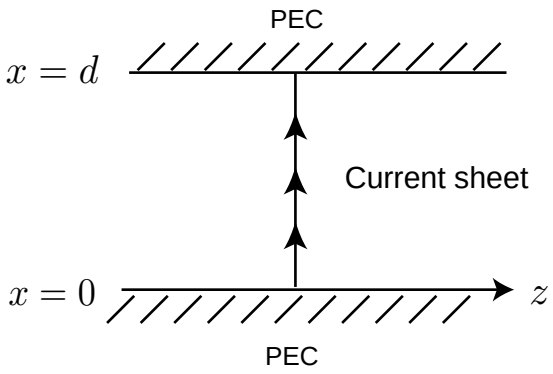
(a) Give the range of frequencies for which only the dominant mode, TE_{10} , may propagate.
(b) Explain why AM signals cannot received.
(c) Can FM(88-108 MHz) signals be received? Above what frequencies?

Problem P10.2

Consider the excitation of a parallel-plate waveguide by a current sheet with

$$\overline{J}_s = \hat{x} J_s \cos \frac{3\pi x}{d}$$

The plates are at $x = 0$ and $x = d$ and the propagation direction is $\hat{z}$. Find the amplitudes of the excited modes. (Hint: Only TM modes are excited in this case.)

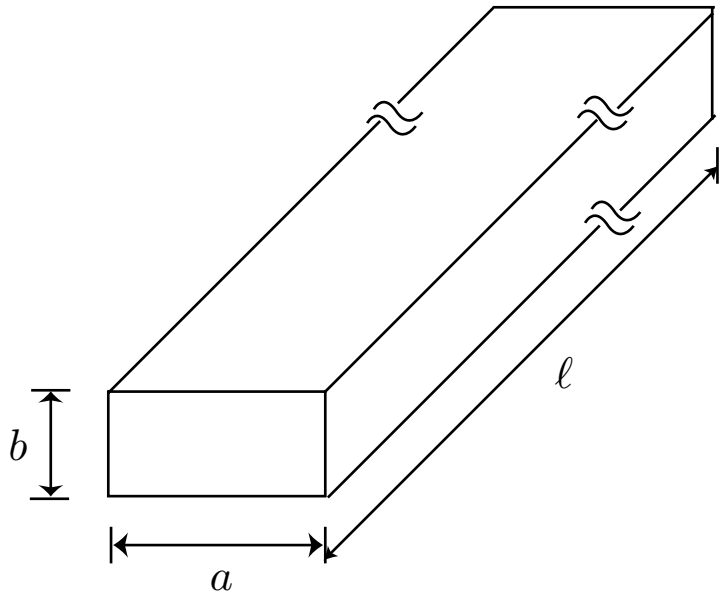


Problem P10.3

A waveguide is often designed such that only the fundamental mode is able to propagate. The reason is that two different propagating modes travel with different velocities, which leads to the difficulty of receiving the clear signals at the other end of the waveguide. Consider a rectangular waveguide as shown in the figure, where $a = 2b = 3$ cm and the length $\ell = 300$ m. It is known that signals travel with their group velocity.

(a) If a signal is modulated at 6 GHz and travels from one end of this waveguide to the other end, how much time will it take?

(b) If a signal is modulated at 10.5 GHz, how many propagating modes can carry this signal? How much time will it take for each of these modes to reach the other end of the waveguide?



MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 8	6.630 Electromagnetics	Issued: 20081030R
Fall Term 2008		Due: 20081106R

Reading assignment: Section 4.1(D-G) ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P8.1

That the exhausted travelers in desert often see an unreal "lake" in the distance is called the phenomenon of mirage. The hot sand in desert heats the air close to the ground and thus forms a layer of air with less density as well as less refractive index. The refractive index of air in the visible region is about 1.000271374 at the temperature of $20°C$, while about 1.000257559 at $35°C$. Consider the two air layers with different temperatures as shown in Fig. 1, what is the critical angle θ_c of the wave in this case? Try to explain the mirage using this model.

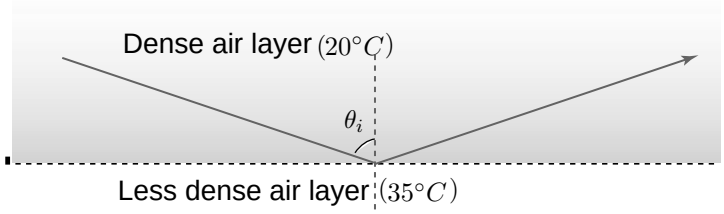


Fig.1

Problem P8.2

Sun light glares caused by reflections from plane surfaces are partially linearly polarized.

(a) Determine the Brewster angle for $\epsilon_t = 9\epsilon_0$ The Brewster angle, θ_B, is also called the polarization angle because at θ_B the reflected wave is entirely perpendicularly polarized (TE polarized).

(b) Your polaroid glasses absorb one linear component of incident light. To minimize sun glare, what component, perpendicularly or parallelly polarized (TM polarized), reaches your eyes after passing through the glasses? Explain why.

Problem P8.3

By comparing the phenomena of total reflection for $\theta > \theta_C$ and total transmission for $\theta = \theta_B$ at an isotropic dielectric interface, we can draw the following conclusions. (1) Total reflection occurs at a range of incident angles larger than the critical angle θ_C ; total transmission of TM waves occurs only at the Brewster angle θ_B . (2) Total reflection occurs only when the incident medium is denser than the transmitted medium. The Brewster angle occurs for any two media. (3) When an unpolarized wave is totally reflected, the reflected wave is still unpolarized. When the TM wave components of an unpolarized wave are totally transmitted, the reflected wave contains only TE waves.

Suppose a TM wave is incident at an angle θ such that $\theta = \theta_B > \theta_C$. Then the wave is totally transmitted and at the same time it is totally reflected. Is that possible? Explain.

Problem P8.4

The gas laser depicted in Fig. 2 uses "Brewster angle" quartz windows on the gas discharge tube in order to minimize reflection losses. Determine the angle θ if the index of refraction for quartz at the wavelength of interest is $n = 1.46$. Because of these windows, the laser output is almost completely linearly polarized. What is the direction of polarization, i.e., is $\overline{E}$ parallel or perpendicular to the paper? Why?

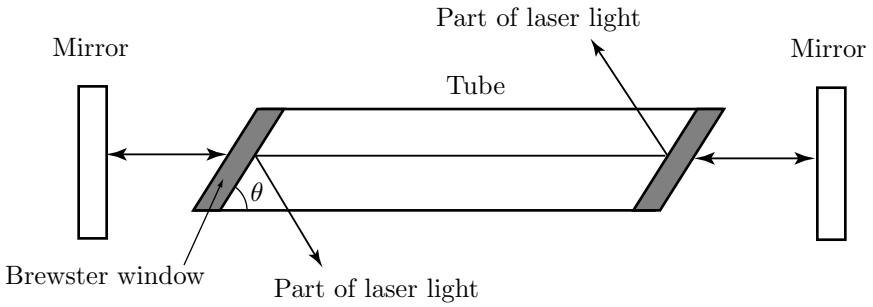


Fig. 2. A gas laser with Brewster windows.

Problem P8.5

Consider a plane wave incident from a dielectric region with permittivity $\epsilon = 3\epsilon_o$ upon a halfspace with $\epsilon = \epsilon_o$ as shown in Fig. 3.

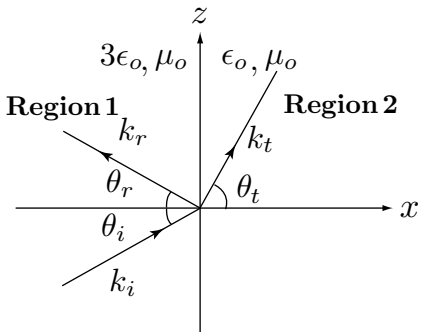


Fig. 3.

(a) Find the Brewster angle for Region 1.

(b) Suppose the transmitted electric field is given by

$$\overline{E}_t = \hat{y}\frac{E_0}{\sqrt{2}}e^{ik_{tx}x+ik_{kz}z} + E_0\frac{\hat{z}-\hat{x}\sqrt{3}}{2\sqrt{2}}e^{ik_{tx}x+ik_{tz}z-i\frac{\pi}{2}}$$

(i) Determine the incident and transmitted angles, θ_i and θ_t.

(ii) What is the polarization of the transmitted field? Be sure to specify the handedness (left or right) if necessary.

(iii) What is the polarization of the reflected wave? Be sure to specify the handedness (left or right) if necessary.

(iv) Give an expression for the incident electric field, $\overline{E}_i$, and the reflected electric field, $\overline{E}_r$.

(c) For this part, assume that the incident wave is a TE wave with magnitude, $|\overline{E}| = E_0$ and $\theta_i = \theta_c$ where θ_c is the critical angle.

(i) Find the Goos-Hänchen phase shift of the reflected wave.

(ii) Make a sketch of the electric field amplitude, E_y, for $x > 0$.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 5	6.632 Electromagnetic Wave Theory	Issued: 20090305 R
Spring Term 2009		Due: 20090312 R

Reading assignment: Section 4.2, 5.1, 6.1C ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing, 2008.

Problem P5.1

Consider an asymmetric slab waveguide shown in Figure 1. It is known that $\epsilon_1 > \epsilon_2 > \epsilon_0$. Derive the guidance condition for TM modes. What is the cutoff spatial frequency for TM$_m$ mode?

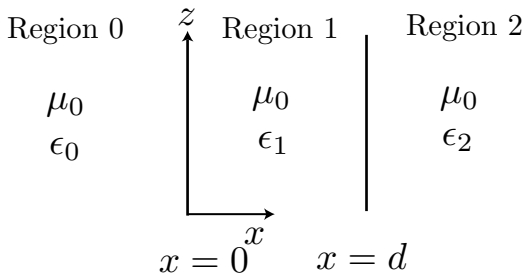


Fig. 1

Problem P5.2

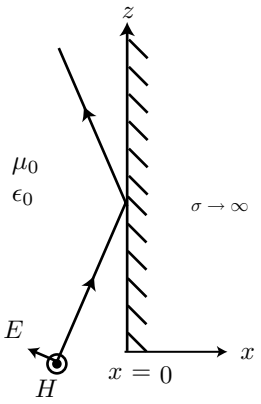


Fig. 2

A plane wave with expression $\overline{E} = (-\hat{x}\frac{\sqrt{3}}{2} + \hat{z}\frac{1}{2})e^{ik_0x/2+ik_0z\sqrt{3}/2}$ which is incident onto a PEC (perfect electric conductor) boundary at $x = 0$ will induce surface current on the interface.

(a) What is the expression of the surface current $\overline{J}_s$?

(b) From another point of view, we can treat the surface current on the interface as a source which generates two waves in $+x$ and $-x$ directions, respectively. The $-x$ travelling wave corresponds to the reflected wave while the $+x$ travelling wave cancels the incident wave in the region of $x > 0$ such that there is no field inside PEC. Show that the radiated waves from the surface current obeys the above statement.

Problem P5.3

In the Čerenkov radiation, the total energy radiated out of a cylinder of path ℓ and radius ρ is given by

$$S_\rho = \frac{\mu q^2 \ell}{4\pi} \int_0^\infty d\omega\, \omega(1 - \frac{1}{n^2\beta^2}).$$

So the energy lost per unit length per unit frequency band is

$$\frac{d^2 S_\rho}{d\ell d\omega} = \frac{\mu q^2}{4\pi}\omega(1 - \frac{1}{n^2\beta^2}).$$

(a) By $E_{\text{photon}} = \hbar\omega$ and $d\omega/d\lambda = 2\pi c/\lambda^2$, show that the number of photons radiated on unit path at wavelength λ is

$$\frac{d^2 N}{d\ell d\lambda} = \frac{q^2 c}{2\lambda^2\hbar}\mu(1 - \frac{1}{n^2\beta^2})$$

and show the frequently used formula $\frac{dN}{d\ell} \propto \frac{d\lambda}{\lambda^2}\sin^2\theta$, which gives the dependence of N on λ and θ.

(b) Gas Čerenkov detector is widely used in high energy particle experiments. The refractive index of the gas n is typically 1.002. What will be the angle for the Čerenkov radiation in case of $\beta = 1$?

(c) Most energy is radiated by the waves in the band 350 nm $\sim$ 550 nm. How many photons can you get on unit path? In order to get 100 photons for the detector, how long is the path (ℓ)? This is the size the detector should be. Note that the parameters are as follows: $\hbar = 6.63 \times 10^{-34}/2\pi\,\text{J}\cdot\text{s/rad}$, $q = 1.6 \times 10^{-19}\,\text{C}$, $\beta = 1$.